

2024-10-22-Convexity

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Slide 1

Content

Convex Analysis (LVJ Chapter 10)

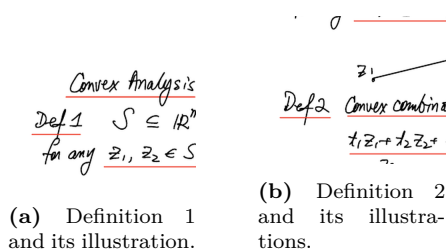
Def 1 $S \subseteq \mathbb{R}^m$ is convex if for any $z_1, z_2 \in S$, $\implies tz_1 + (1-t)z_2 \in S$ $0 \leq t \leq 1$

Def 2 Convex combination of $z_1, z_2, \dots, z_n \in \mathbb{R}^m$ $t_1z_1 + t_2z_2 + \dots + t_nz_n$ $t_i \geq 0$ $\sum_i t_i = 1$

Description

This slide introduces the concepts of convex sets and convex combinations in the context of convex analysis, specifically chapter 10 of a textbook by LVJ. Definition 1 states that a set S in $\mathbb{R}^m$ is convex if, for any two points z_1 and z_2 in S , the line segment connecting them (represented by the convex combination $tz_1 + (1-t)z_2$ where $0 \leq t \leq 1$) is entirely contained within S . A simple illustration shows two points z_1 and z_2 and the line segment between them. Definition 2 defines a convex combination of n points $z_1, z_2, \dots, z_n$ in $\mathbb{R}^m$ as a weighted sum $t_1z_1 + t_2z_2 + \dots + t_nz_n$, where the weights t_i are non-negative and sum to 1. The illustrations show examples of convex combinations for two and three points, forming a line segment and a triangle respectively. The context of linear programming is relevant because convexity is a crucial property of feasible regions in linear programming problems.

Figures



Slide 2

Content

Thm 10.1 S is convex $\iff$ it contains all conv. comb. of points in S pf $\iff$

Suppose S contain all conv. comb. of pts in S Then clearly, for any $z_1, z_2 \in S$ $tz_1 + (1-t)z_2 \in S$ Conv. comb. of $z_1, z_2 \implies \underline{S \text{ is convex}}$

Description

This slide presents Theorem 10.1, which states that a set S is convex if and only if it contains all convex combinations of its points. The proof focuses on one direction of the implication: if S contains all convex combinations of its points, then S is convex. The proof begins by assuming that S contains all convex combinations of its points. Then, it shows that for any two points z_1 and z_2 in S , their convex combination $tz_1 + (1-t)z_2$ (where $0 \leq t \leq 1$) is also in S by the assumption. This directly satisfies the definition of a convex set, thus proving that S is convex. The underlining highlights the key assumption and conclusion of this part of the proof. The notation "pf" indicates "proof," and the double-headed arrow indicates the "if and only if" nature of the theorem. "conv. comb." is an abbreviation for "convex combination". The context of linear programming is crucial because this theorem is fundamental to understanding the properties of feasible regions in linear programming problems, which are often convex sets.

Figures

Thm 10.1 S is a
if contains all its
pf $\iff$

(a) Theorem 10.1
statement and
proof outline.

Slide 3

Content

Thm 10.1 S is convex $\iff$ it contains all conv. comb. of points in S pf $\implies$ Suppose S is convex.

$$n = 2: z_1, z_2 \in S \implies t_1 z_1 + t_2 z_2 \in S, t_1, t_2 \geq 0, t_1 + t_2 = 1$$

$$n = 3: z_1, z_2, z_3 \in S \implies t_1 z_1 + t_2 z_2 + t_3 z_3 = (t_1 + t_2) \left(\frac{t_1}{t_1 + t_2} z_1 + \frac{t_2}{t_1 + t_2} z_2 \right) + t_3 z_3, t_1 + t_2 + t_3 = 1$$

$$t_1, t_2 \geq 0, t_3 \geq 0 \implies t_1 z_1 + t_2 z_2 + t_3 z_3 \in S$$

Description

This slide presents Theorem 10.1 and outlines a proof by induction showing that if a set S is convex, then it contains all convex combinations of its points. The theorem statement asserts the equivalence between the convexity of S and the inclusion of all convex combinations of points in S . The proof starts by assuming S is convex. The base case ($n = 2$) demonstrates that if z_1 and z_2 are in S , then any convex combination $t_1 z_1 + t_2 z_2$ (with $t_1, t_2 \geq 0$ and $t_1 + t_2 = 1$) is also in S due to the definition of convexity. The inductive step ($n = 3$) shows that if $z_1, z_2, z_3 \in S$, then a convex combination $t_1 z_1 + t_2 z_2 + t_3 z_3$ can be rewritten as a convex combination of a convex combination of z_1 and z_2 and z_3 . Since the inner convex combination is in S (by the base case), and the outer convex combination is also in S (by the definition of convexity), the entire expression is in S . This inductive argument can be extended to any number of points. The underlining highlights the key assumption. The notation "pf" stands for "proof," and "conv. comb." is short for "convex combination." The context of linear programming is crucial because this theorem is fundamental to understanding the properties of feasible regions in linear programming problems, which are often convex sets. The implication arrow ($\implies$) indicates the direction of the proof.

Figures

Thm 10.1 S is a
it contains all its
pf " $\implies$ " Suppose S
 $n=2; z_1, z_2 \in S \implies$

(a) Theorem 10.1
statement and
proof outline for
 $n=2$ and $n=3$.

Slide 4

Content

Thm 10.1 S is convex $\iff$ it contains all conv. comb. of points in S pf $\implies$ Suppose S is convex.

$n = 2$: $z_1, z_2 \in S \implies t_1 z_1 + t_2 z_2 \in S, t_1, t_2 \geq 0, t_1 + t_2 = 1$

$n = 4$: $z_1, z_2, z_3, z_4 \in S$ $t_1 z_1 + t_2 z_2 + t_3 z_3 + t_4 z_4 = (t_1 + t_2 + t_3) \left(\frac{t_1}{t_1 + t_2 + t_3} z_1 + \frac{t_2}{t_1 + t_2 + t_3} z_2 + \frac{t_3}{t_1 + t_2 + t_3} z_3 \right) + t_4 z_4$ $(t_1 + t_2 + t_3) + t_4 = 1, t_1, t_2, t_3 \geq 0, t_4 \geq 0 \implies t_1 z_1 + t_2 z_2 + t_3 z_3 + t_4 z_4 \in S$

Description

This slide continues the proof of Theorem 10.1 by induction, showing that if S is a convex set, then it contains all convex combinations of its points. The previous slide covered the base case ($n = 2$) and the inductive step for $n = 3$. This slide extends the inductive step to $n = 4$. It assumes that $z_1, z_2, z_3, z_4 \in S$. The convex combination $t_1 z_1 + t_2 z_2 + t_3 z_3 + t_4 z_4$ is rewritten as a convex combination of a convex combination of z_1, z_2, z_3 and z_4 . Because the inner convex combination is in S (by the inductive hypothesis, assuming the result holds for $n = 3$), and the outer convex combination is also in S (by the definition of convexity), the entire expression is in S . This demonstrates the inductive step, implying the theorem holds for all n . The underlining highlights the key assumption. The notation "pf" stands for "proof," and "conv. comb." is short for "convex combination." The context of linear programming is crucial because this theorem is fundamental to understanding the properties of feasible regions in linear programming problems, which are often convex sets. The implication arrow ($\implies$) indicates the direction of the proof. The handwritten notes show a slight error in the constraints on the t_i values; they should all be non-negative and sum to 1.

Figures

Thm 10.1 S is a
it contains all its
pf " $\implies$ " Suppose S
 $n=2; z_1, z_2 \in S \implies$

(a) Theorem 10.1
statement and
proof outline for
 $n=2$ and $n=4$.

Slide 5

Content

Def $S \subseteq \mathbb{R}^m$, Convex Hull of S ,

$$\text{Conv}(S) = \bigcap G$$

G convex, $S \subseteq G$

(= Smallest convex set containing S)

(Note intersection of convex sets is convex.)

Thm 10.2 $S \subseteq \mathbb{R}^m$, $\text{Conv}(S)$ = collection of
all conv. comb. of finitely many pts from S

Description

This slide introduces the definition of the convex hull of a set and presents Theorem 10.2. The definition states that for a set S in $\mathbb{R}^m$, the convex hull of S , denoted as $\text{Conv}(S)$, is the intersection of all convex sets containing S . It is explicitly stated that this is equivalent to the smallest convex set containing S . A note is added that the intersection of convex sets is itself a convex set. Theorem 10.2 then states that for a set S in $\mathbb{R}^m$, the convex hull of S is equal to the collection of all convex combinations of finitely many points from S . The underlining highlights the definition and the theorem statement. The notation " $\text{Conv}(S)$ " represents the convex hull of set S , and "conv. comb." is an abbreviation for "convex combination". The context of linear programming is crucial because the convex hull is a fundamental concept in understanding the feasible region of a linear program, which is always a convex set. The theorem provides a practical way to compute or characterize the convex hull.

Figures

$$\begin{aligned} \text{Def } S \subseteq \mathbb{R}^m, \text{ Conv} \\ \text{Conv}(S) &= \bigcap \\ &\quad G \text{ convex} \\ &= \text{Smallest} \end{aligned}$$

(a) Definitions of
Convex Hull and
Theorem 10.2.

Slide 6

Content

Thm 10.2 $S \subseteq \mathbb{R}^m$, $\text{Conv}(S)$ = collection of
 all conv. comb. of finitely many pts from S
pf. Let H = collection of all conv. comb.
 of finitely many pts from S
 Need to show $\text{Conv}(S) = H$
 (1) $\text{Conv}(S) \subseteq H$ ($\because H$ is conv.)

Description

This slide presents Theorem 10.2 and outlines the beginning of its proof. Theorem 10.2 states that for a set S in $\mathbb{R}^m$, the convex hull of S , denoted $\text{Conv}(S)$, is equal to the set of all convex combinations of finitely many points from S . The proof starts by defining H as the set of all convex combinations of finitely many points from S . The goal is to show that $\text{Conv}(S) = H$. The slide shows the first part of the proof, aiming to demonstrate that $\text{Conv}(S)$ is a subset of H . The underlining highlights this key step in the proof. The notation " $\text{Conv}(S)$ " represents the convex hull of set S , "conv. comb." is short for "convex combination", and "pts" is short for "points". The implication ($\because$) indicates that the inclusion follows because H is a convex set. The context of linear programming is crucial because this theorem provides a practical way to compute or characterize the convex hull of a feasible region, which is essential in solving linear programming problems.

Figures

Thm 10.2 $S \subseteq \mathbb{R}^m$, $\text{Conv}(S)$ =
 all conv. comb. of
 pf. Let H = collection
 of finitely many
 pts from S
 Need to show $\text{Conv}(S) = H$
 (1) $\text{Conv}(S) \subseteq H$ ($\because H$ is conv.)

(a) Theorem 10.2
 statement and
 proof outline.

Slide 8

Content

Caratheodory Theorem

THEOREM 10.3. The convex hull $\text{conv}(S)$ of a set S in $\mathbb{R}^m$ consists of all convex combinations of $m + 1$ points from S :

$$\text{conv}(S) = \left\{ z = \sum_{j=1}^{m+1} t_j z_j : z_j \in S \text{ and } t_j \geq 0 \text{ for all } j, \text{ and } \sum_j t_j = 1 \right\}$$

$$\text{pf } \text{conv}(S) = \left\{ z = \sum_{j=1}^n x_j z_j : z_j \in S, x_j \geq 0, \sum_j x_j = 1 \right\}$$

Consider the LP: $\max c^T x$

s.t. $Ax = z$

$\sum x_i = 1$

$x_i \geq 0$

(z is given, fixed, $A = [z_1, z_2, \dots, z_n]_{m \times n}$)

Description

This slide presents Caratheodory's Theorem (Theorem 10.3) and outlines the beginning of its proof within the context of linear programming. The theorem states that the convex hull of a set S in $\mathbb{R}^m$, denoted $\text{conv}(S)$, consists of all convex combinations of $m + 1$ points from S . The proof begins by restating the definition of the convex hull as the set of all convex combinations of points from S , but with the key difference that it uses only n points instead of $m + 1$. The proof then introduces a linear program (LP) to find a convex combination of points in S that equals a given point z . The LP maximizes $c^T x$ subject to $Ax = z$, where A is a matrix whose columns are points from S , and x is a vector of coefficients representing the convex combination. The constraint $\sum x_i = 1$ ensures that the combination is convex, and $x_i \geq 0$ ensures non-negativity of the coefficients. The underlining highlights the theorem statement and the key equation in the proof. The notation " $\text{conv}(S)$ " represents the convex hull of set S , and the subscript $m \times n$ indicates the dimensions of matrix A . The context of linear programming is crucial because the proof uses an LP to demonstrate the theorem. The theorem itself is important because it provides a bound on the number of points needed to represent any point in the convex hull, which has implications for computational efficiency in linear programming algorithms.

Figures

Caratheodory Theorem

THEOREM 10.3. The convex hull $\text{conv}(S)$ consists of convex combinations of $m + 1$ points from S :

$$\text{conv}(S) = \left\{ z = \sum_{j=1}^{m+1} t_j z_j : z_j \in S \text{ and } t_j \geq 0, \sum_{j=1}^{m+1} t_j = 1 \right\}$$

pf $\text{conv}(S) = \left\{ z = \sum_{j=1}^n x_j z_j \right\}$

(a) Caratheodory Theorem statement and proof outline.

Slide 9

Content

Caratheodory Theorem

THEOREM 10.3. The convex hull $\text{conv}(S)$ of a set S in $\mathbb{R}^m$ consists of all convex combinations of $m + 1$ points from S :

$$\text{conv}(S) = \left\{ z = \sum_{j=1}^{m+1} t_j z_j : z_j \in S \text{ and } t_j \geq 0 \text{ for all } j, \text{ and } \sum_j t_j = 1 \right\}$$

pf $\text{conv}(S) = \left\{ z = \sum_{j=1}^n x_j z_j : z_j \in S, x_j \geq 0, \sum_j x_j = 1 \right\}$

Consider the LP: $\max c^T x$

s.t. $Ax = z$

$\sum x_i = 1$

$x_i \geq 0$

$\left\{ \begin{array}{l} m+1 \\ \text{constraints} \end{array} \right\}$

This is feasible, and hence has a basic opt. soln

with $m + 1$ basic variables.

Description

This slide presents Caratheodory's Theorem (Theorem 10.3) and outlines the beginning of its proof. The theorem states that the convex hull of a set S in $\mathbb{R}^m$ consists of all convex combinations of at most $m + 1$ points from S . The proof starts by noting that the convex hull can be defined as the set of all convex combinations of points from S . Then, a linear program (LP) is introduced to find a convex combination of points in S that equals a given point z . The LP maximizes $c^T x$ subject to $Ax = z$, where A is a matrix whose columns are points from S , and x is a vector of coefficients representing the convex combination. The constraint $\sum x_i = 1$ ensures that the combination is convex, and $x_i \geq 0$ ensures non-negativity of the coefficients. The handwritten note points out that this LP has $m + 1$ constraints and is feasible, therefore it has a basic optimal solution with $m + 1$ basic variables. This implies that only $m + 1$ points from S are needed to represent z , proving the theorem. The underlining highlights the theorem statement and the key equation in the proof. The notation " $\text{conv}(S)$ " represents the convex hull of set S , and the subscript $m \times n$ indicates the dimensions of matrix A . The context of linear programming is crucial because the proof uses an LP to demonstrate the theorem. The theorem itself is important because it provides a bound on the number of points needed to represent any point in the convex hull, which has implications for computational efficiency in linear programming algorithms.

Figures

Caratheodory Theorem

THEOREM 10.3. The convex hull $\text{conv}(S)$ consists of all convex combinations of $m+1$ points from S :

$$\text{conv}(S) = \left\{ z = \sum_{j=1}^{m+1} t_j z_j : z_j \in S \text{ and } t_j \geq 0 \text{ for all } j, \text{ and } \sum_j t_j = 1 \right\}$$

pf $\text{conv}(S) = \left\{ z = \sum_{j=1}^n x_j z_j \right\}$

(a) Caratheodory Theorem statement and proof outline.

Slide 10

Content

Separation Theorem

THEOREM 10.4. Let P and $\bar{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\bar{H}$ such that $P \subset H$ and $\bar{P} \subset \bar{H}$.

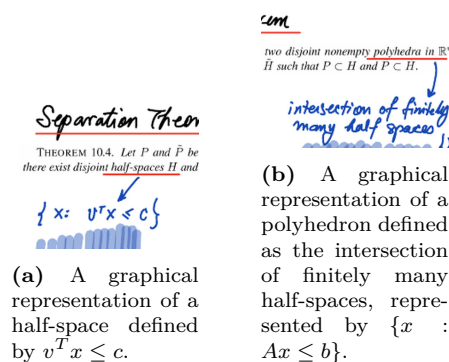
↓

$$\{x : v^T x \leq c\} \quad \{x : Ax \leq b\}$$

Description

This slide introduces the Separation Theorem (Theorem 10.4) in the context of linear programming. The theorem states that if P and $\bar{P}$ are two disjoint nonempty polyhedra in $\mathbb{R}^n$, then there exist two disjoint half-spaces, H and $\bar{H}$, such that P is a subset of H and $\bar{P}$ is a subset of $\bar{H}$. The slide visually represents this concept. The left figure shows a half-space, which is represented mathematically as $\{x : v^T x \leq c\}$, where v is a vector and c is a scalar. The right figure depicts a polyhedron, which is shown as the intersection of multiple half-spaces, represented mathematically as $\{x : Ax \leq b\}$, where A is a matrix and b is a vector. The arrow points to the mathematical representation of a polyhedron as the intersection of finitely many half-spaces. The theorem is fundamental in linear programming because it provides a way to separate feasible and infeasible regions, which is crucial for developing algorithms to solve linear programming problems. The visual aids help to understand the geometric interpretation of the theorem.

Figures



Slide 11

Content

Separation Theorem

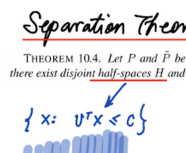
THEOREM 10.4. Let P and $\bar{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\bar{H}$ such that $P \subset H$ and $\bar{P} \subset \bar{H}$.

$\{x : v^T x \leq c\}$ intersection of finitely many half spaces $\{x : Ax \leq b\}$

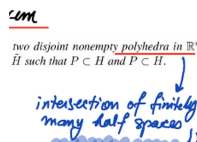
Description

This slide presents the Separation Theorem (Theorem 10.4) which states that given two disjoint nonempty polyhedra, P and $\bar{P}$, in $\mathbb{R}^n$, there exist two disjoint half-spaces, H and $\bar{H}$, such that P is contained in H and $\bar{P}$ is contained in $\bar{H}$. The theorem is illustrated graphically. The left figure depicts a half-space, $\{x : v^T x \leq c\}$, represented as a shaded region bounded by a line. The right figure shows a polyhedron, $\{x : Ax \leq b\}$, represented as a shaded region formed by the intersection of multiple half-spaces. The handwritten text clarifies that a polyhedron is the intersection of finitely many half-spaces. The drawing shows two polyhedra, P and $\bar{P}$, separated by a line (a hyperplane in higher dimensions), which represents the separating half-spaces H and $\bar{H}$. The arrows indicate the correspondence between the graphical representation and the mathematical notation. The theorem is fundamental in linear programming because it guarantees the existence of a separating hyperplane between disjoint feasible and infeasible regions, which is crucial for the development of algorithms to solve linear programming problems. The context of linear programming is essential for understanding the significance of this theorem in optimization problems.

Figures



(a) A graphical representation of a half-space defined by $v^T x \leq c$. The figure shows a shaded region representing the half-space, bounded by a line. An arrow points to the mathematical representation of the half-space.



(b) A graphical representation of a polyhedron defined as the intersection of finitely many half-spaces, represented by $\{x : Ax \leq b\}$. The figure shows a shaded polyhedral region formed by the intersection of several half-spaces. An arrow points to the mathematical representation of the polyhedron. A separating hyperplane is drawn between two polyhedra P and $\bar{P}$.

Slide 12

Content

Separation Theorem

THEOREM 10.4. Let P and $\tilde{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\tilde{H}$ such that $P \subset H$ and $\tilde{P} \subset \tilde{H}$.

$$P \neq \emptyset \quad P = \{x : Ax < b\}, \quad \tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$$

$$P \cap \tilde{P} = \emptyset \iff \begin{cases} Ax < b \\ \tilde{A}x < \tilde{b} \end{cases} \text{ has no solution}$$

$\iff$ There is a $y, \tilde{y}$ s.t.

$$y^T A + \tilde{y}^T \tilde{A} = 0$$

$$y^T b + \tilde{y}^T \tilde{b} < 0$$

$$y, \tilde{y} \geq 0$$

Description

This slide presents the Separation Theorem (Theorem 10.4) and its proof using Farkas' Lemma. The theorem states that two disjoint nonempty polyhedra, P and $\tilde{P}$, in $\mathbb{R}^n$ can be separated by disjoint half-spaces H and $\tilde{H}$. The polyhedra are defined as $P = \{x : Ax < b\}$ and $\tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$. The core of the slide is the equivalence: the intersection of P and $\tilde{P}$ is empty if and only if the system of inequalities $\{Ax < b, \tilde{A}x < \tilde{b}\}$ has no solution. This is then shown to be equivalent to the existence of vectors y and $\tilde{y}$ satisfying the conditions $y^T A + \tilde{y}^T \tilde{A} = 0$, $y^T b + \tilde{y}^T \tilde{b} < 0$, and $y, \tilde{y} \geq 0$. This last part is a direct application of Farkas' Lemma, a fundamental result in linear programming. The notation "<" represents strict inequality, and the superscript T denotes the transpose of a vector or matrix. The context of linear programming is crucial because the theorem and its proof are fundamental to the theory of linear programming, particularly in duality theory and the development of algorithms.

Figures

Separation Theorem

Theorem 10.4. Let P and $\tilde{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\tilde{H}$ such that $P \subset H$ and $\tilde{P} \subset \tilde{H}$.

$$\begin{aligned} P &= \{x : Ax < b\}, \quad \tilde{P} = \{x : \tilde{A}x < \tilde{b}\} \\ P \cap \tilde{P} = \emptyset &\iff \begin{cases} Ax < b \\ \tilde{A}x < \tilde{b} \end{cases} \text{ has no solution} \end{aligned}$$

$\iff$ There is a $y, \tilde{y}$ s.t.

(Farkas' Lemma) $\begin{cases} y^T A + \tilde{y}^T \tilde{A} = 0 \\ y^T b + \tilde{y}^T \tilde{b} < 0 \\ y, \tilde{y} \geq 0 \end{cases}$

(a) Handwritten notes outlining the Separation Theorem and its proof using Farkas' Lemma.

Slide 13

Content

Separation Theorem

THEOREM 10.4. Let P and $\tilde{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\tilde{H}$ such that $P \subset H$ and $\tilde{P} \subset \tilde{H}$.

$$P \neq \emptyset \quad P = \{x : Ax < b\}, \quad \tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$$

$$P \cap \tilde{P} = \emptyset \iff \begin{cases} Ax < b \\ \tilde{A}x < \tilde{b} \end{cases} \text{ has no solution}$$

Farkas' Lemma

LEMMA 10.5. The system $Ax \leq b$ has no solutions if and only if there is a y such that

$$(10.8) \quad A^T y = 0 \iff \begin{pmatrix} y^T A = 0 \\ y \geq 0 \\ y^T b < 0 \end{pmatrix}$$

Description

This slide presents the Separation Theorem (Theorem 10.4) and connects it to Farkas' Lemma (Lemma 10.5). The Separation Theorem states that two disjoint nonempty polyhedra, P and $\tilde{P}$, in $\mathbb{R}^n$ can be separated by disjoint half-spaces. The polyhedra are defined using strict inequalities: $P = \{x : Ax < b\}$ and $\tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$. The slide then shows that the intersection $P \cap \tilde{P}$ is empty if and only if the system of inequalities $\{Ax < b, \tilde{A}x < \tilde{b}\}$ has no solution. This is where Farkas' Lemma comes in. Farkas' Lemma (Lemma 10.5) provides a condition for when a system of inequalities $Ax \leq b$ has no solution. It states that there is no solution if and only if there exists a vector y such that $A^T y = 0$, $y \geq 0$, and $y^T b < 0$. The slide neatly presents this equivalence using a matrix notation on the right. The handwritten notes show the connection between the Separation Theorem and Farkas' Lemma, indicating that the non-existence of a solution to the combined system of inequalities from the Separation Theorem can be determined using Farkas' Lemma. The number (10.8) suggests this is equation 10.8 in the textbook or lecture notes. The context of linear programming is crucial because both theorems are fundamental to the theory of linear programming, particularly in duality theory and the development of algorithms. The use of strict inequalities ($<$) instead of weak inequalities ($\leq$) in the definition of the polyhedra is a subtle but important detail.

Figures

Separation Theorem

THEOREM 10.4. Let P and $\tilde{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\tilde{H}$ such that $P \subset H$ and $\tilde{P} \subset \tilde{H}$.

$$P = \{x : Ax < b\}, \quad \tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$$

$$P \cap \tilde{P} = \emptyset \iff \begin{cases} Ax < b \\ \tilde{A}x < \tilde{b} \end{cases} \text{ has no solution}$$

Farkas' Lemma

LEMMA 10.5. The system $Ax \leq b$ has no solutions if and only if there is a

$$(10.8) \quad \begin{pmatrix} A^T y = 0 \\ y \geq 0 \\ y^T b < 0 \end{pmatrix}$$

(a) Handwritten notes outlining the Separation Theorem and its proof using Farkas' Lemma.

Slide 14

Content

Separation Theorem

THEOREM 10.4. Let P and $\tilde{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\tilde{H}$ such that $P \subset H$ and $\tilde{P} \subset \tilde{H}$.

$$P \neq \emptyset \quad P = \{x : Ax < b\}, \quad \tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$$

$$\text{There is a } y, \tilde{y} \text{ s.t.} \quad \begin{cases} y^T A + \tilde{y}^T \tilde{A} = 0 \\ y^T b + \tilde{y}^T \tilde{b} < 0 \\ y, \tilde{y} \geq 0 \end{cases}$$

Define

$$H := \{x : y^T A x < y^T b\}, \quad \tilde{H} := \{x : \tilde{y}^T \tilde{A} x < \tilde{y}^T \tilde{b}\}$$

$$H \cap \tilde{H} = \emptyset, \quad P \subset H, \quad \tilde{P} \subset \tilde{H}$$

Description

This slide presents the Separation Theorem (Theorem 10.4) and outlines a proof. The theorem states that two disjoint nonempty polyhedra, P and $\tilde{P}$, in $\mathbb{R}^n$ can be separated by disjoint half-spaces H and $\tilde{H}$. The polyhedra are defined as $P = \{x : Ax < b\}$ and $\tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$, using strict inequalities. The core of the slide is the assertion that there exist vectors y and $\tilde{y}$ satisfying the conditions $y^T A + \tilde{y}^T \tilde{A} = 0$, $y^T b + \tilde{y}^T \tilde{b} < 0$, and $y, \tilde{y} \geq 0$. These conditions are not explicitly proven on the slide but are likely derived from Farkas' Lemma (as seen in the previous slide). The slide then defines the separating half-spaces H and $\tilde{H}$ using these vectors y and $\tilde{y}$. The final statement asserts that the intersection of H and $\tilde{H}$ is empty, and that P is a subset of H and $\tilde{P}$ is a subset of $\tilde{H}$. The notation "<" represents strict inequality, and the superscript T denotes the transpose. The context of linear programming is crucial because this theorem is fundamental to the theory of linear programming, particularly in duality theory and the development of algorithms. The use of strict inequalities (<) instead of weak inequalities ($\leq$) in the definition of the polyhedra is a subtle but important detail.

Figures

Separation Theorem
Verification: (10.4). Let P and $\tilde{P}$ be two disjoint nonempty polyhedra in $\mathbb{R}^n$. Then there exist disjoint half-spaces H and $\tilde{H}$ such that $P \subset H$ and $\tilde{P} \subset \tilde{H}$.

$$\text{Let } P = \{x : Ax < b\}, \quad \tilde{P} = \{x : \tilde{A}x < \tilde{b}\}$$

$$\text{There is a } y, \tilde{y} \text{ s.t.} \quad \begin{cases} y^T A + \tilde{y}^T \tilde{A} = 0 \\ y^T b + \tilde{y}^T \tilde{b} < 0 \\ y, \tilde{y} \geq 0 \end{cases}$$

$$\text{Define } H := \{x : y^T A x < y^T b\}, \quad \tilde{H} := \{x : \tilde{y}^T \tilde{A} x < \tilde{y}^T \tilde{b}\}.$$

$$H \cap \tilde{H} = \emptyset, \quad P \subset H, \quad \tilde{P} \subset \tilde{H}$$

(a) Handwritten notes outlining the Separation Theorem and its proof.

Slide 15

Content

Farkas' Lemma

LEMMA 10.5. The system $Ax \leq b$ has no solutions if and only if there is a y such that

$$\begin{aligned} A^T y &= 0 \\ y &\geq 0 \\ b^T y &< 0 \end{aligned}$$

(10.8)

PROOF. Consider the linear program

$$\begin{aligned} &\text{maximize } 0 \\ &\text{subject to } Ax \leq b \end{aligned}$$

and its dual

$$\begin{aligned} &\text{minimize } b^T y \\ &\text{subject to } A^T y = 0 \\ &\quad y \geq 0 \end{aligned}$$

Description

This slide presents Farkas' Lemma (Lemma 10.5), a fundamental result in linear programming. The lemma states that the system of inequalities $Ax \leq b$ has no solution if and only if there exists a vector y satisfying the conditions $A^T y = 0$, $y \geq 0$, and $b^T y < 0$. The proof outline begins by considering a primal linear program that maximizes 0 subject to $Ax \leq b$. The dual of this linear program is then introduced, which minimizes $b^T y$ subject to $A^T y = 0$ and $y \geq 0$. The proof likely proceeds by showing the equivalence between the primal problem having no feasible solution and the dual problem being unbounded (or having an objective function value less than 0). This connection between the primal and dual problems is a key element of linear programming duality theory. The equation number (10.8) suggests this is equation 10.8 in the textbook or lecture notes. The context of linear programming is crucial because Farkas' Lemma is a cornerstone of the theory, used extensively in proving other important results, such as the strong duality theorem and the separation theorem.

Figures

Farkas' Lemma.

LEMMA 10.5. The system $Ax \leq b$ has no solutions if and only if there is

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(10.8)

PROOF. Consider the linear program

$$\begin{aligned} &\text{maximize } 0 \\ &\text{subject to } Ax \leq b \end{aligned}$$

and its dual

$$\begin{aligned} &\text{minimize } b^T y \\ &\text{subject to } A^T y = 0 \\ &\quad y \geq 0. \end{aligned}$$

(a) Handwritten notes outlining Farkas' Lemma and its proof using linear programming duality.